Characterizing Limb Tremor as a Multi-Degree of Freedom (MDOF) System Involving Springs and Dampers



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Limb tremor is commonly linked to ET and Parkinson's, with millions affected nationally

Common Causes of Limb Tremor

Essential Tremor (ET)

- Most common cause of action tremor
- Exact cause is unknown
- Happens during movement – writing, eating

Parkinson's Disease

- Caused by loss of dopamine-producing brain cells
- Tremor strongest at rest

Tremor Band

- The range of frequencies that tremor tend to occur
- 2-12 Hz

Scale of problem in the U.S

6.38M ~ 7.63M

Estimated Essential Tremor Cases in the USA in 2012

2.2 %

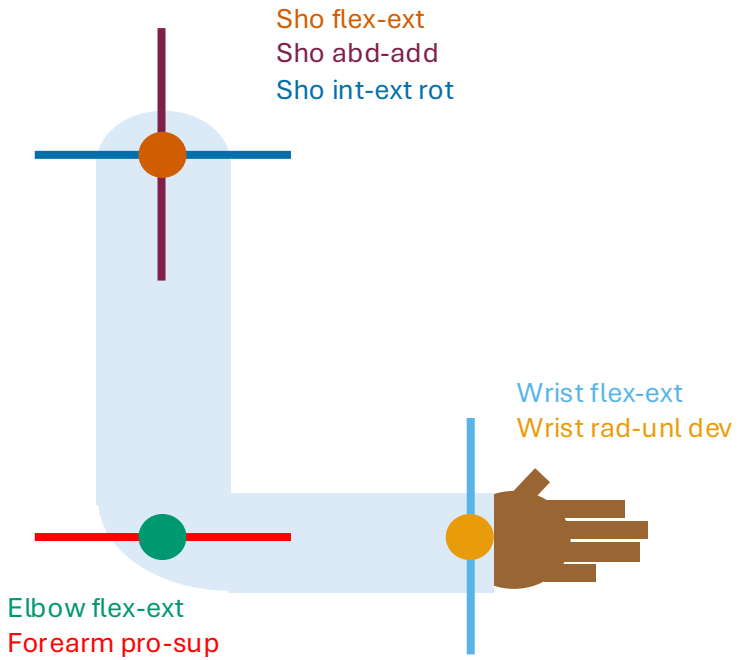
Of the US population

90,000

People in the U.S are estimated to be diagnosed with Parkinson's

Model Setup | Equation of Motion

7-DOF System Setup and Assumptions



- Linear, time-invariant
- Constant posture
- Input Tremor acts in one DOF at a time

Equation of Motion

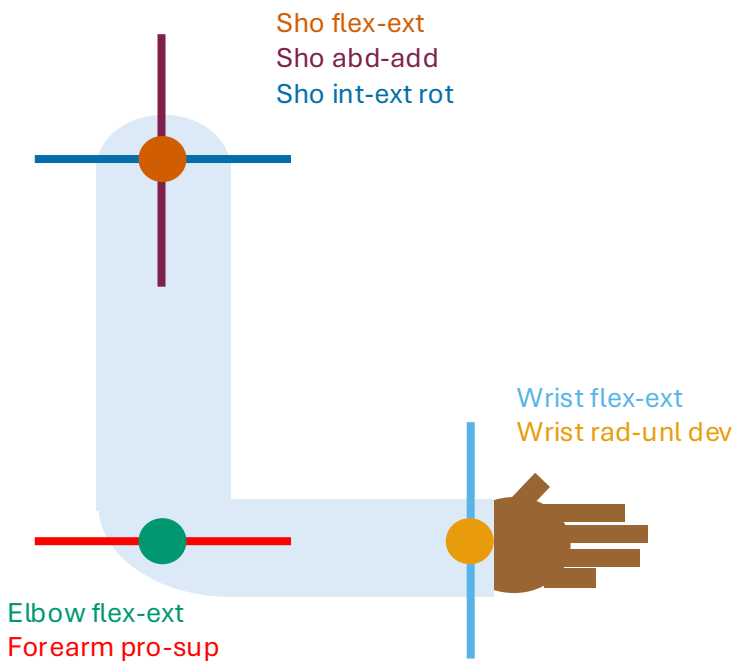
$$I\ddot{q} + R\dot{q} + kq = \tau$$

Symbol	Name	Units
I	Inertia Matrix	$kg\ m^2$
R	Rotational Damping Matrix	$\frac{N\ m\ s}{rad}$
K	Rotational stiffness Matrix	$\frac{N\ m}{rad}$
τ	Tremor Torque	$N\ m$

Where, $\tau(t) = Asin(w_t t)$

Model Setup | Rotational Stiffness Matrix (K)

System Sketch

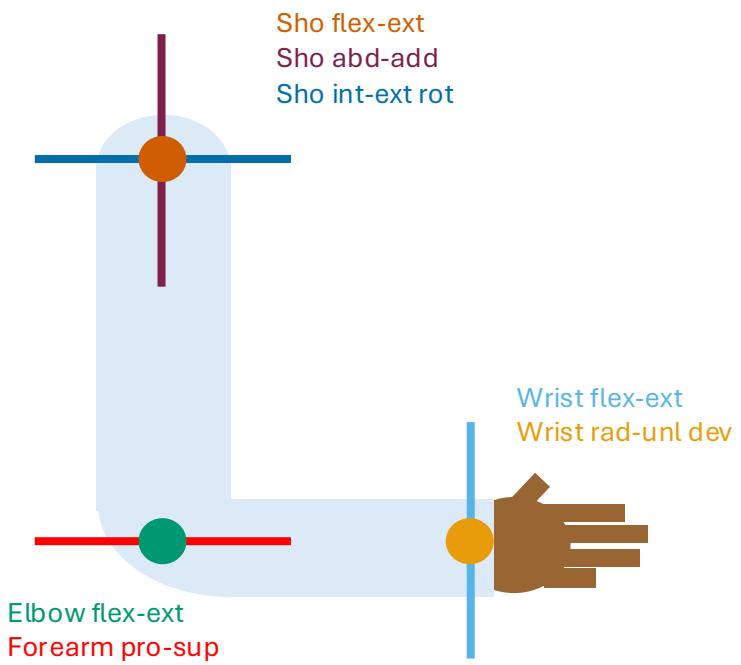


Rotational Stiffness Matrix (Nms/rad)

	Sho flex-ext	Sho abd-add	Sho int-ex rot	Elbow fex-ext	Forearm pro-sup	Wrist flex-ext	Wrist rad-unl dev
Sho flex-ext	10.80	2.626	0.279	2.670	0.000	0.000	0.000
Sho abd-add	2.626	5.468	3.821	0.000	0.000	0.000	0.000
Sho int-ex rot	0.279	3.821	7.486	0.000	0.000	0.000	0.000
Elbow fex-ext	2.670	0.000	0.000	8.670	0.000	0.000	0.000
Forearm pro-sup	0.000	0.000	0.000	0.000	0.756	0.018	0.291
Wrist flex-ext	0.000	0.000	0.000	0.000	0.018	0.992	-0.099
Wrist rad-unl dev	0.000	0.000	0.000	0.000	0.291	0.099	2.920

Model Setup | Rotational Damping Matrix (R)

System Sketch

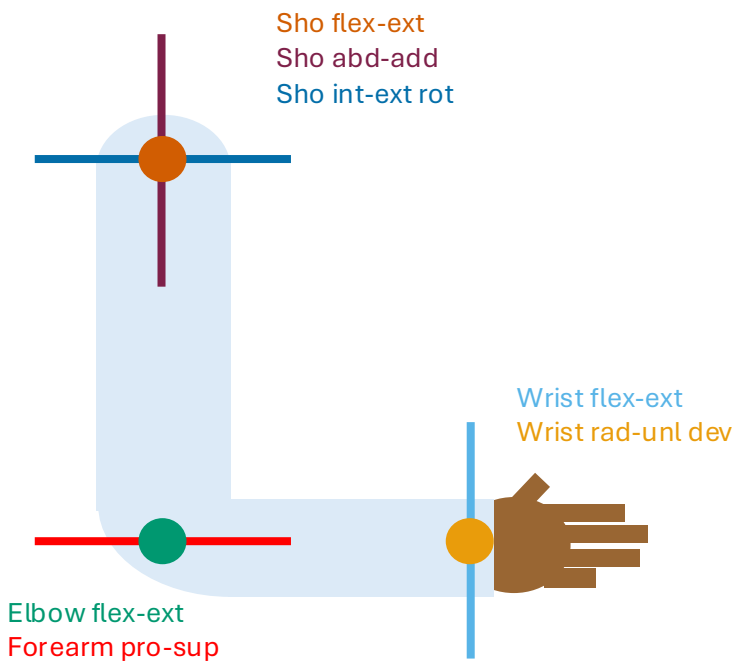


Rotational Damping Matrix (Nm/rad)

	Sho flex-ext	Sho abd-add	Sho int-ex rot	Elbow fex-ext	Forearm pro-sup	Wrist flex-ext	Wrist rad-unl dev
Sho flex-ext	0.756	0.184	0.020	0.187	0	0	0
Sho abd-add	0.184	0.383	0.267	0	0	0	0
Sho int-ex rot	0.020	0.267	0.524	0	0	0	0
Elbow fex-ext	0.187	0	0	0.607	0	0	0
Forearm pro-sup	0	0	0	0	0.021	0.001	0.008
Wrist flex-ext	0	0	0	0	0.001	0.028	-0.003
Wrist rad-unl dev	0	0	0	0	0.008	-0.003	0.082

Model Setup | Inertial Matrix (I)

System Sketch

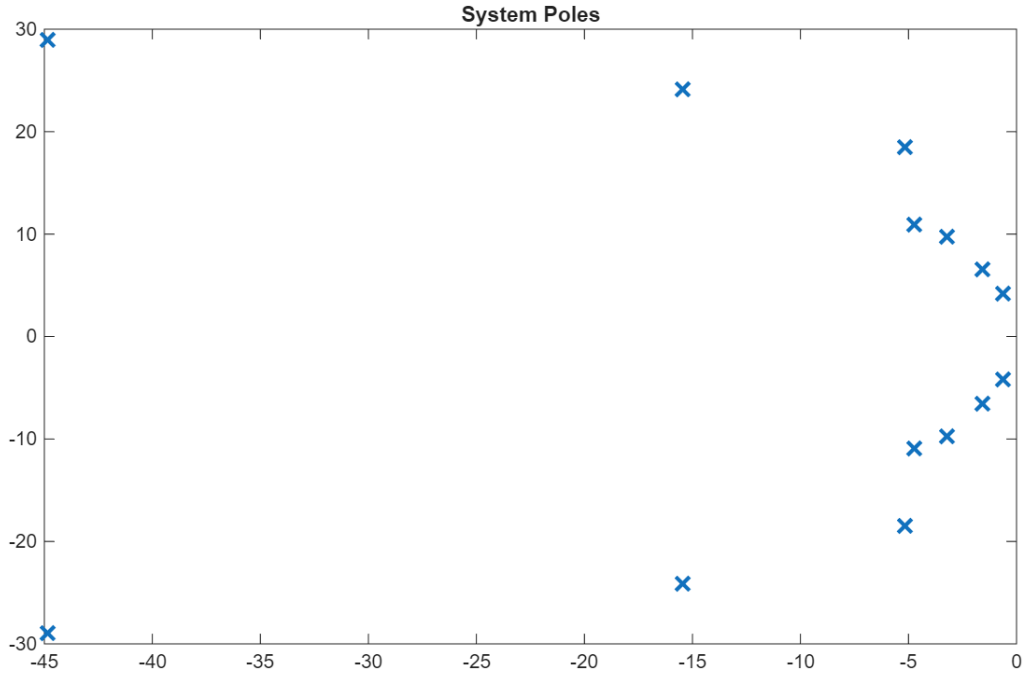


Inertial Matrix (kgm^2)

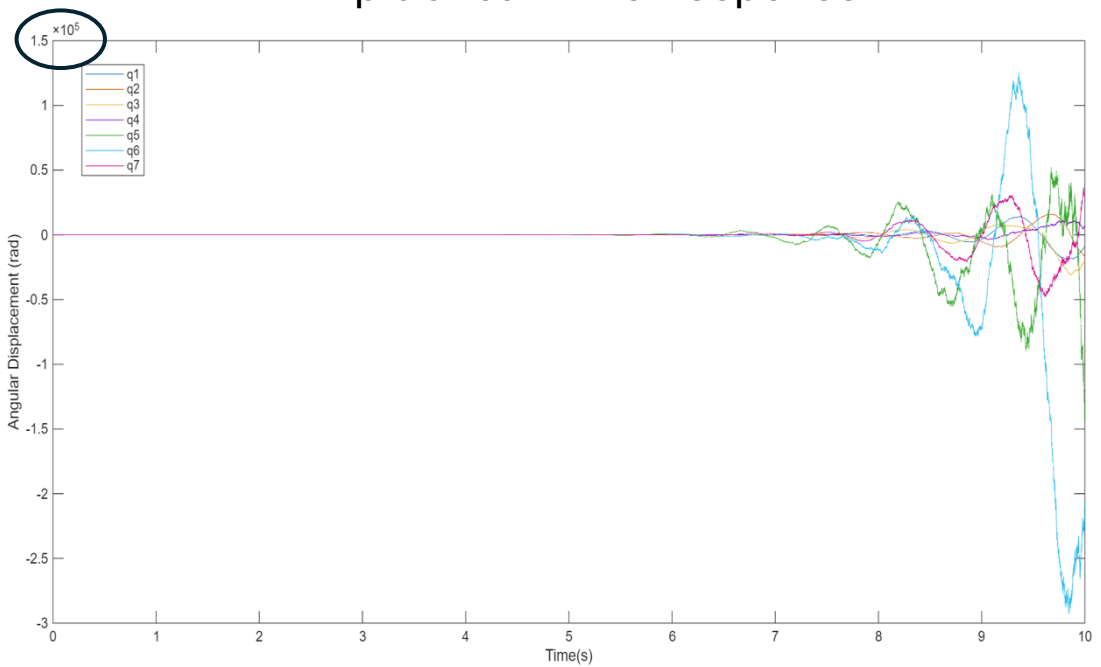
	Sho flex-ext	Sho abd-add	Sho int-ex rot	Elbow fex-ext	Forearm pro-sup	Wrist flex-ext	Wrist rad-unl dev
Sho flex-ext	0.269	0	0	0.076	0	0	-0.014
Sho abd-add	0	0.196	0.083	0	-0.002	0.009	0
Sho int-ex rot	0	0.083	0.079	0	0	0.011	0
Elbow fex-ext	0.076	0	0	0.076	0	0	-0.012
Forearm pro-sup	0	-0.002	0	0	0.002	0	0
Wrist flex-ext	0	0.009	0.011	0	0	0.003	0
Wrist rad-unl dev	-0.014	0	0	-0.012	0	0	0.003

Using Laplace methods to examine the time domain responses yielded an erratic response, despite the model being stable.

14 Left Hand Poles



Impractical Time Response



The model was represented in the state space form and the minreal() and balreal() functions applied to the model.

State Space Representation:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

EOM breakdown:

$$I\ddot{q} + R\dot{q} + Kq = \tau \rightarrow \ddot{q} = I^{-1}\tau - I^{-1}R\dot{q} - I^{-1}Kq$$

Matrix breakdown:

$$A = \begin{bmatrix} F & E \\ -I^{-1}K & -I^{-1}R \end{bmatrix}, B = \begin{bmatrix} F \\ I^{-1} \end{bmatrix}, C = [E \quad F], D = [F]$$

$14x14$ $14x7$ $7x14$ $7x7$

Where:

1) $x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$ and $\tau = u$

2) $F = zeros(7,7)$ and $E = eye(7)$

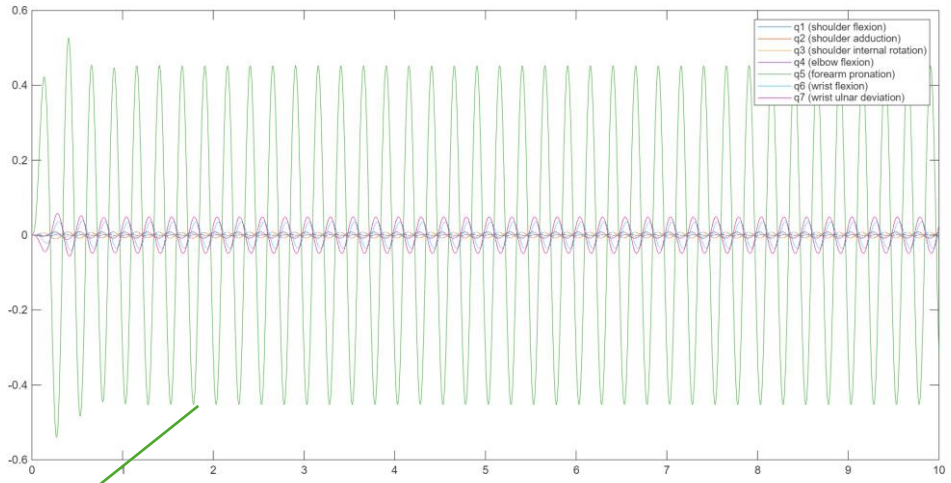
The Resonant Frequencies of the system showed amplification of tremor torques would be highest at the low end of the tremor band ($2 - 5\ Hz$).

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}$$

Pole Number	$\omega_n\ (Hz)$	ζ	$\omega_r\ (Hz)$	
1	0.57	0.147	0.65	
2	1.08	0.235	1.02	
3	1.60	0.269	1.51	
4	1.88	0.313	1.70	
5	3.05	0.398	2.52	Within Tremor Band
6	4.64	0.538	2.96	
7	8.57	0.839	-	

The FPS, WFE, and WRUD DOFs demonstrated the highest degree of angular displacement for an input tremor torque of $\tau = 0.33\sin(8\pi t)$ applied in their DOF.

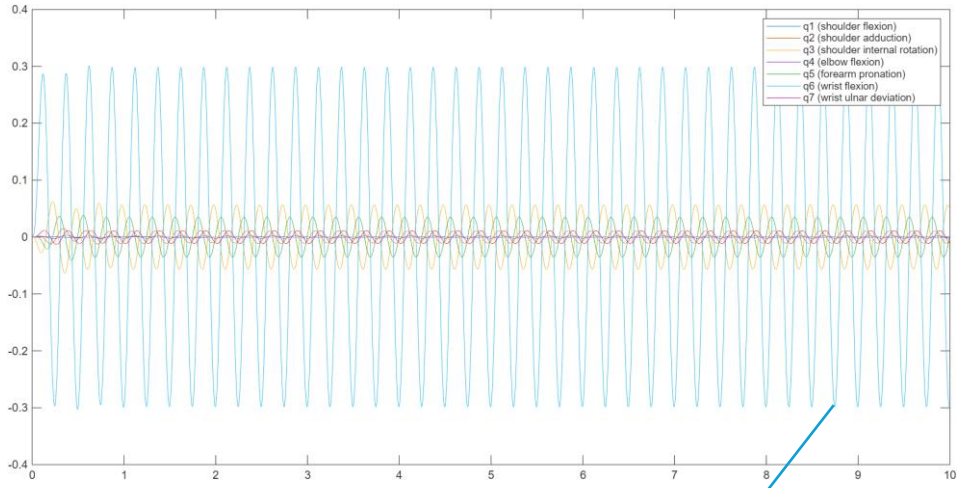
For Input Torque in DOF 5



25°

Severe Tremor

For Input Torque in DOF 6



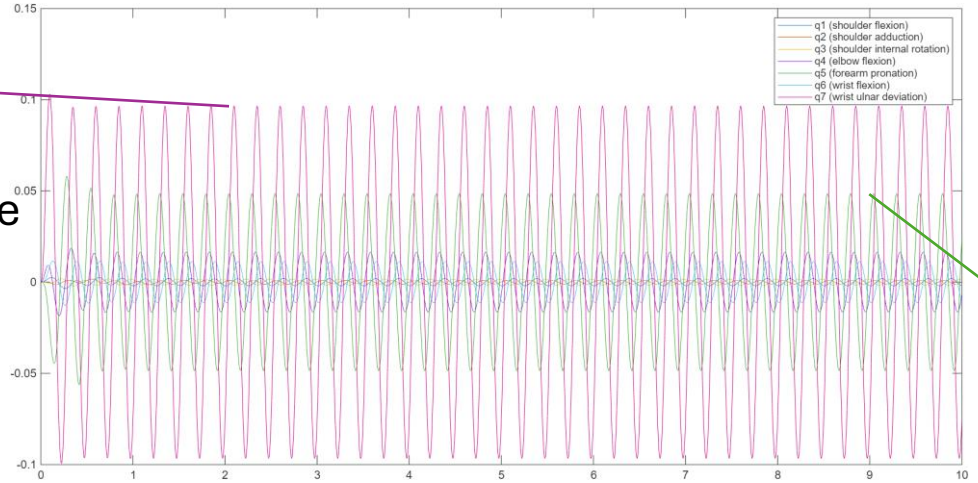
18°

Severe Tremor

For Input Torque in DOF 7

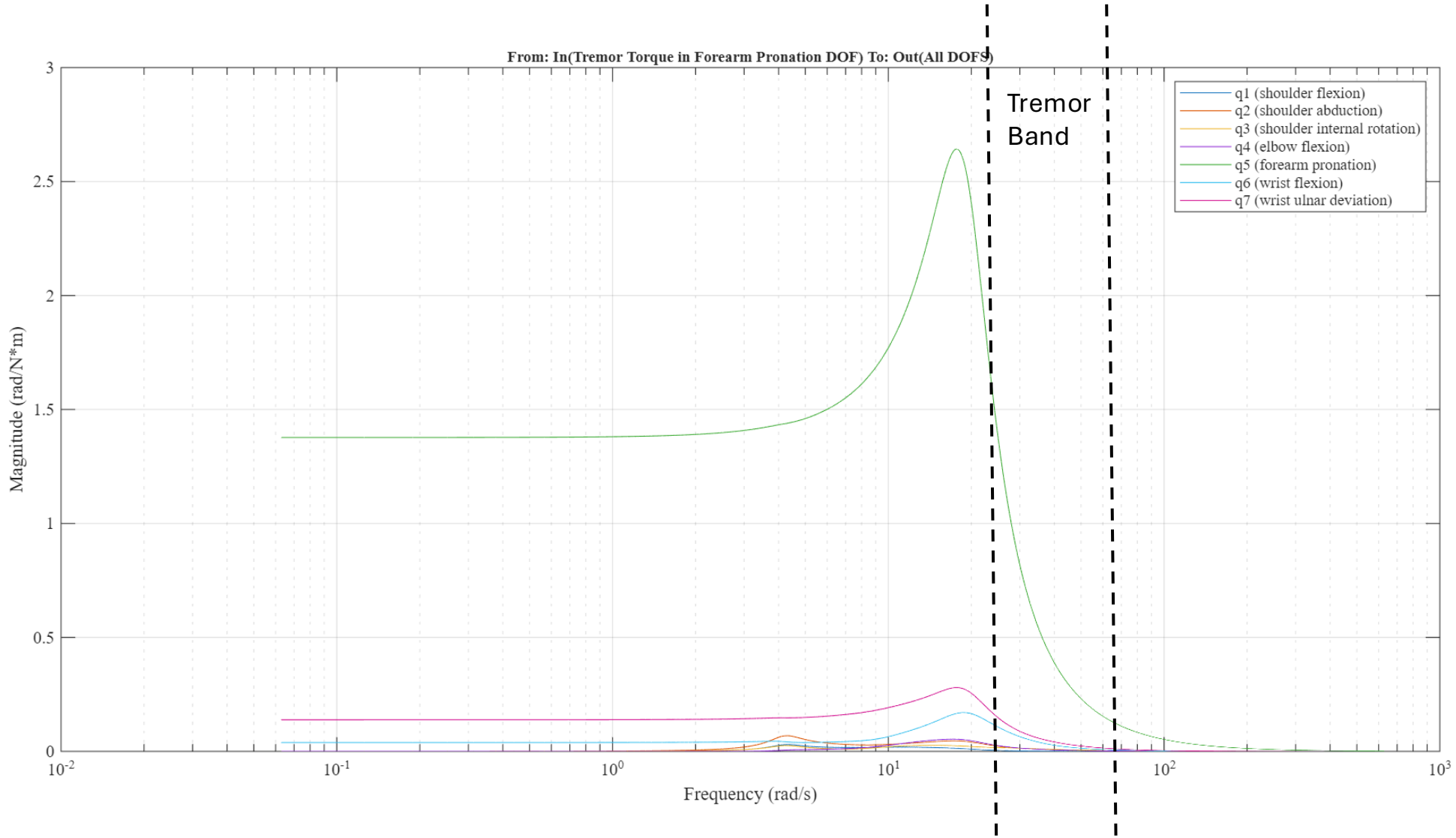
5°

Mild-Moderate Tremor

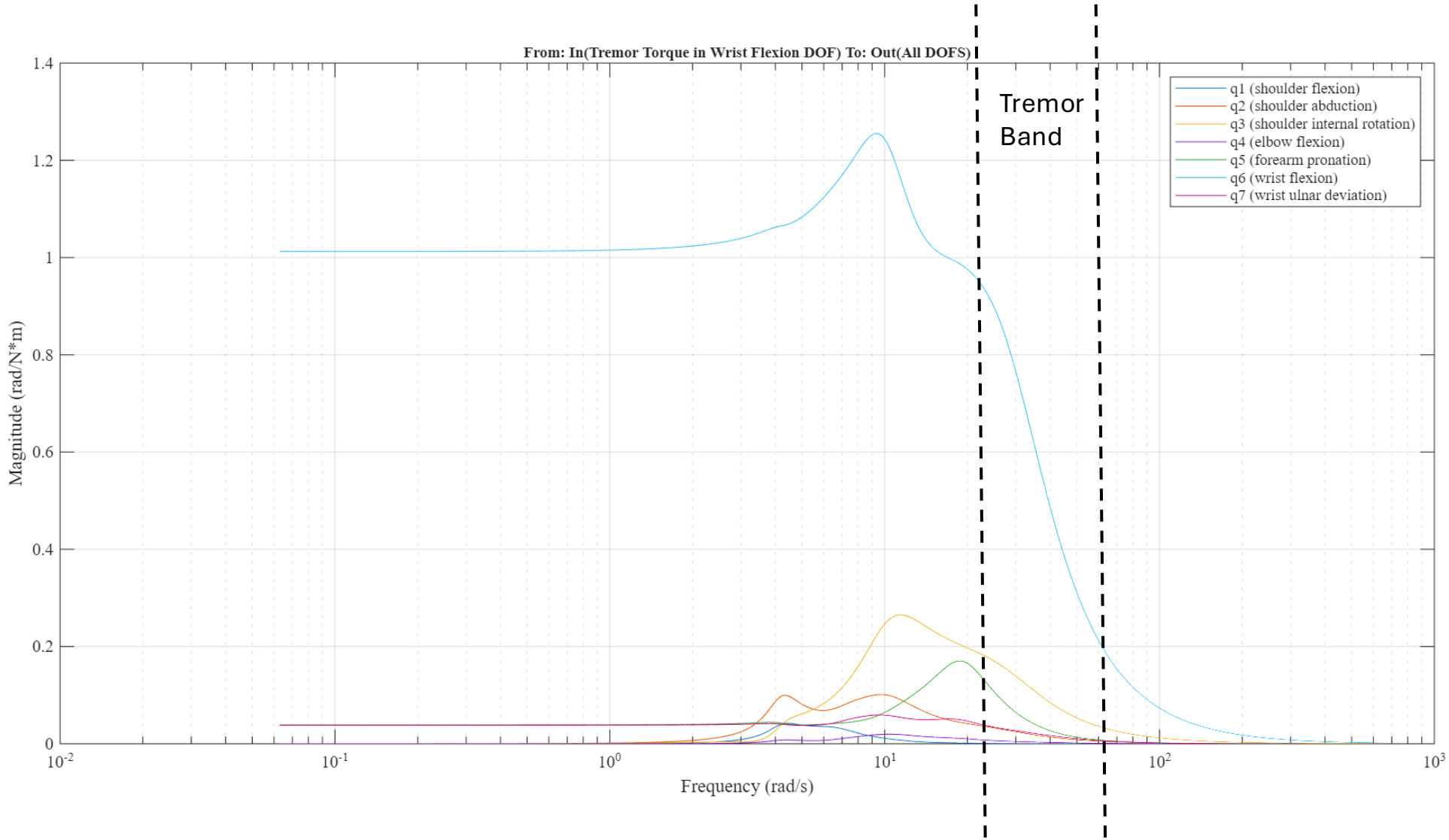


Tremor Propagation from coupling

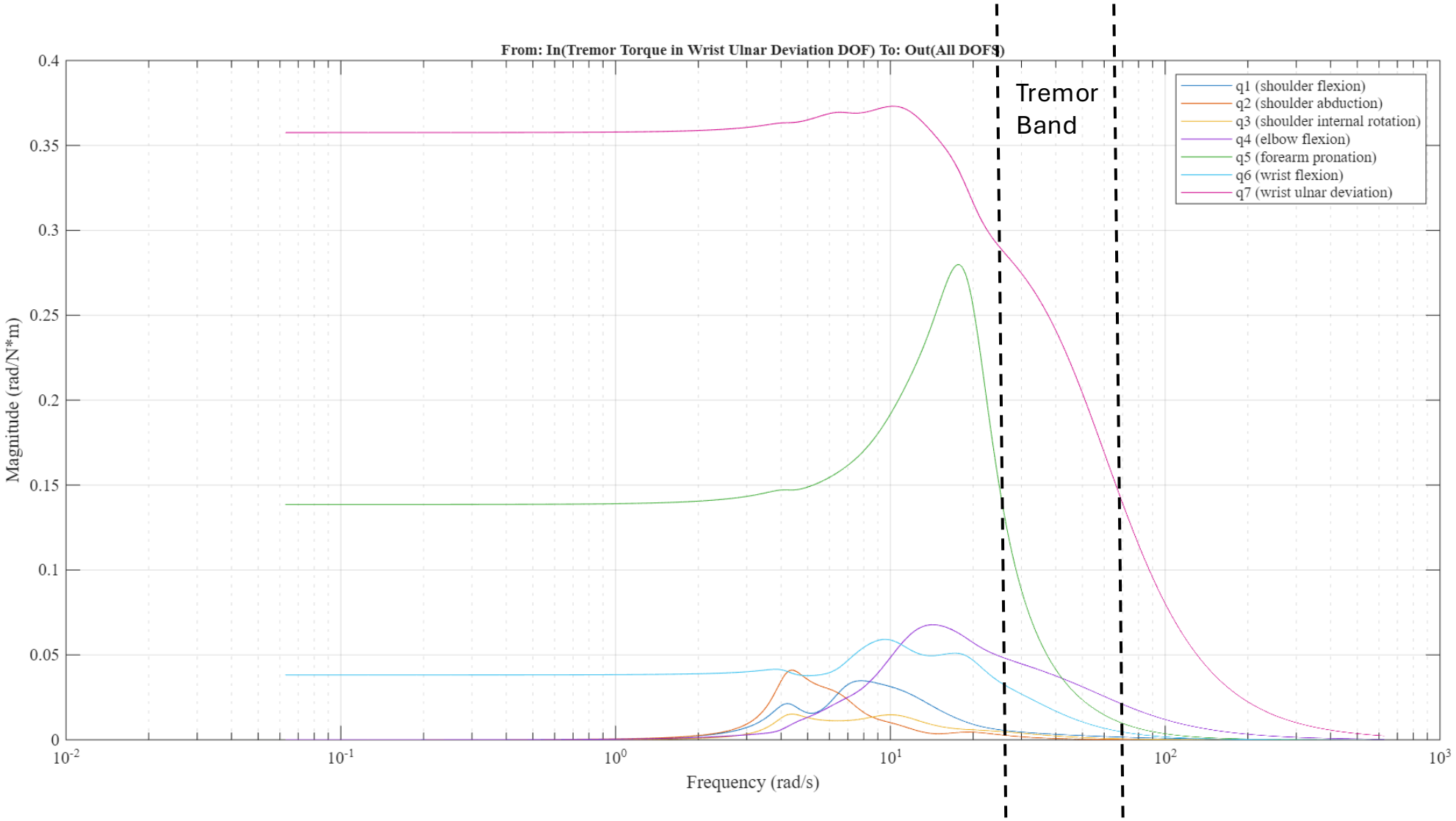
Input tremor in the Forearm Pronation DOF produced the largest magnitude peak, showing that DOF 5 was the most sensitive to tremor.



Input tremor in the Wrist Flexion DOF strongly amplifies its own response which confirmed high distal-joint sensitivity.

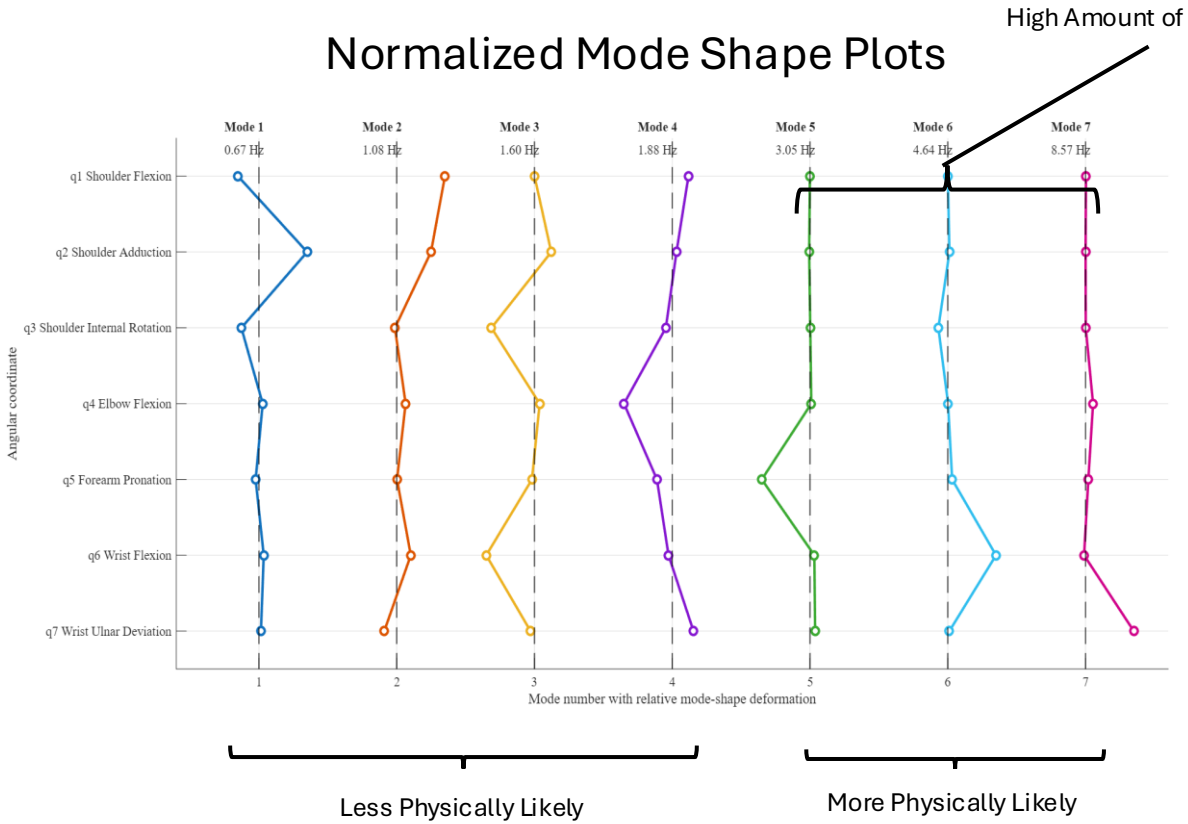


Input tremor in the Wrist Ulnar Deviation DOF produced moderate self-response and noticeable coupling into forearm pronation DOF.

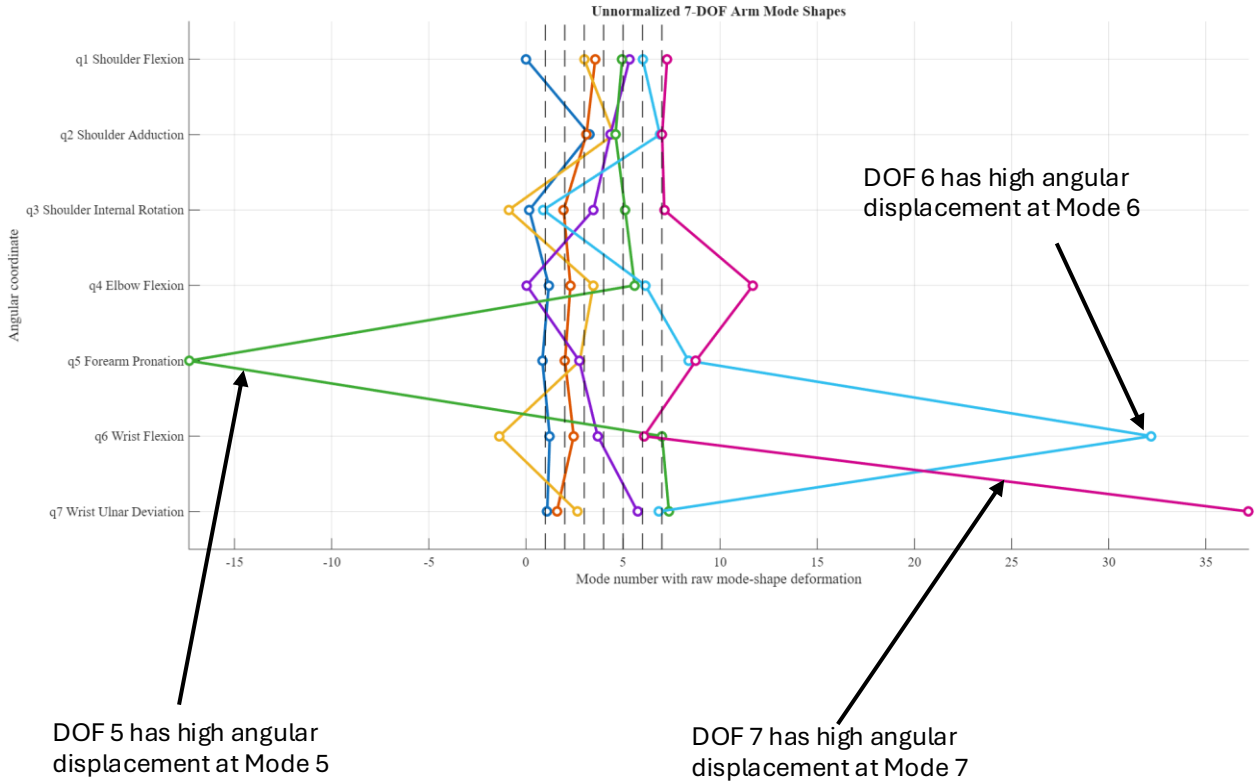


Because Modes 5–7 align most closely with the 2–12 Hz tremor band, they are the most physically relevant modes and likely represent the most severe tremor conditions.

Normalized Mode Shape Plots



Non-normalized Mode Shape Plots



To decrease the tremor in DOF 5-7, a passive weighted wristband was rigidly attached to the wrist and was simulated by adding I_{device} to the original I matrix.

For Diagonal Terms:

$$I_{device\,i,j} = m_r L_i L_j \quad for\ i,j = 1;7\ and\ i = j$$

For Non-diagonal Terms:

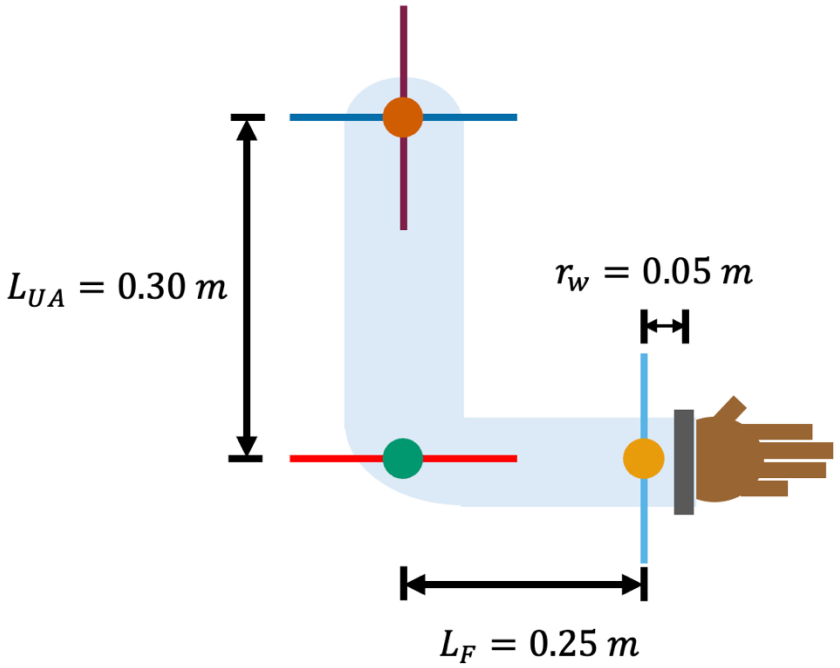
$$I_{device\,i,j} = m_r L_i L_j \cos(\theta_i - \theta_j) \quad for\ i,j = 1;7\ and\ i \neq j$$

Calculated Matrix:

$$I_{device} = \begin{bmatrix} 0.0468 & 0 & 0 & 0 & 0 & 0 & 0.004446 \\ 0 & 0.0468 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0468 & 0 & 0 & 0.004446 & 0 \\ 0 & 0 & 0 & 0.0234 & 0 & 0 & 0.0039 \\ 0 & 0 & 0 & 0 & 0.000416 & 0 & 0 \\ 0 & 0 & 0.004446 & 0 & 0 & 0.0234 & 0 \\ 0.004446 & 0 & 0 & 0.0039 & 0 & 0 & 0.00065 \end{bmatrix}$$

Total Matrix:

$$I_{total} = I_{original} + I_{device}$$



Knowing that:

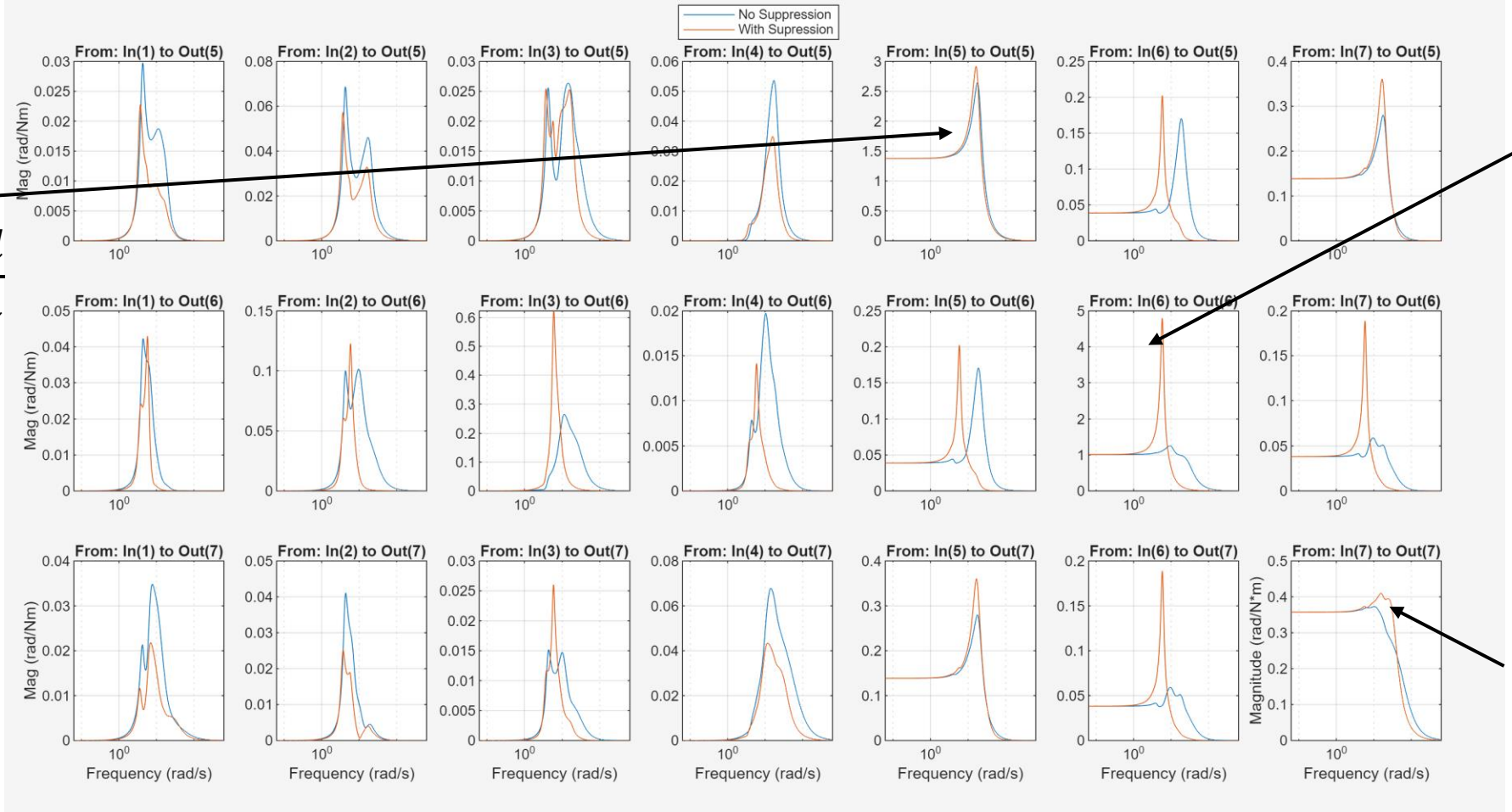
$$\omega_n = \sqrt{\frac{k}{I}}$$

The inclusion of wristband generally shifted the magnitude response leftward, decreasing the movement in DOF 6, but offering little to no change in DOF 5 and 7 at $\omega_t = 4\ Hz$.

DOF 5:
Gain:

$1.34 \rightarrow 1.01 \frac{rad}{Nm}$

Tremor:
$25° \rightarrow 18°$
(Still Severe)



DOF 6:
Gain:

$0.94 \rightarrow 0.02 \frac{rad}{Nm}$

Tremor:
$18° \rightarrow 0.5°$
(None-Mild)

DOF 7:
Gain:

$0.28 \rightarrow 0.39 \frac{rad}{Nm}$

Tremor:
$5° \rightarrow 7°$
(Still Moderate)

The main limitations with the baseline model was that the analysis only applied to a singular posture and tremor input, and that the device was rigidly attached to the wrist.

Modeling assumptions

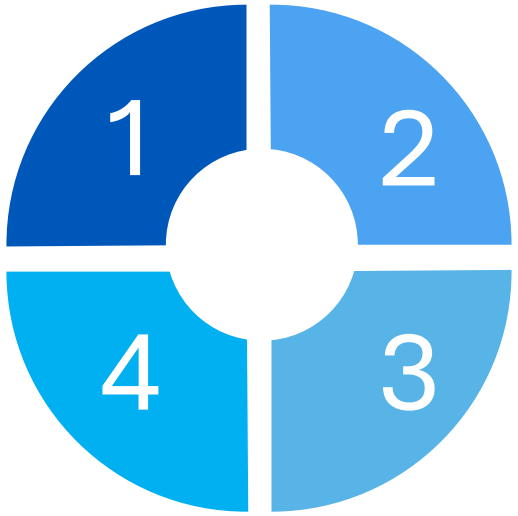
- Linear, time-invariant model:
  - Constant I, R, and K matrices.
  - Nonlinear muscle behavior and changing joint properties not included.
- Fixed posture:
  - Doesn't represent posture changes.
- Same tremor input for all DOFs:
  - Real tremor strength can vary by joint and by person.
- Single main test frequency:
  - Time-domain response focused on 4 Hz tremor input.
  - Behavior across the full tremor band mainly interpreted through FRF.

Device assumptions

- Idealized Modeling:
  - Assumed as rigid, ignoring shifts
- Exclusion of Biological Factors:
  - Doesn't account for voluntary muscle correction/reflexes
- Uneven Joint Suppression:
  - Effectiveness varies by joint (increases tremor displacement in DOF 7)

In the future, the device can be tuned by including damping and more stiffness, and a wearable prototype can be tested in the tremor band.

1. Add coupled stiffness/damping terms to model a real brace
2. Test full tremor band, multiple amplitudes, and multiple postures
3. Build a wearable prototype and validate model with motion/sensor data
4. Optimize I device , K device, and R device



Citations

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Thank you ! Any Questions?

